

MDL Assignment 4

Part A: Probability Theory

Deadline: April 18th 2026

Question 1 [40 Marks]

A farmer has a field and can choose to plant any one of the following three crops each season: **Corn**, **Wheat**, or **Saffron**. Because of the unpredictable weather, there is a risk that a crop might fail.

The probabilities of success are as follows:

- The probability that planted **Corn** is harvested successfully is $x\%$.
- The probability that planted **Wheat** is harvested successfully is $x\%$.
- The probability that **Saffron** is harvested successfully is $(100 - 2x)\%$.

Note: x is always less than 50 ($x < 50$).

Market Payouts:

- If a **Corn** or **Wheat** harvest is successful, the farmer receives $3\times$ the money they spent.
- If a **Saffron** harvest is successful, the farmer receives a massive $10\times$ the money they spent.

The farmer starts with \$1000 and wants to grow this money over 1000 seasons.

Planting Strategies to Test:

1. **The Corn Specialist:** Every season, the farmer spends exactly \$10 to plant Corn.
2. **The "Double-Down" Strategy:** The farmer starts by spending \$10 on Wheat. If the harvest fails, they double their spending the next season to try and recover the loss. If the harvest succeeds, they return to the original \$10 in the next season.
3. **Saffron Is All You Need:** Every season, the farmer spends exactly \$10 to plant Saffron.
4. **The Chaos Method:** Every season, the farmer randomly picks one of the three crops and spends \$10 on it.

Task:

Evaluate these strategies using Monte Carlo simulations for different values of x . You must choose **any 4 values of x** such that $40 < x < 50$. You should simulate **1000 seasons**, and repeat this **1000 times** for each strategy to get a steady average.

Calculate the **expected return** (average final amount of money the farmer has) and the **risk of ruin** (the percentage of times the farmer goes broke and hits \$0). Include an analysis on which strategy is the best for someone who hates losing money and why.

Additional Details:

- Initial capital: \$1000.
- Base cost to plant any crop: \$10.
- **Bankruptcy Rule:** For the "Double-Down" strategy, if the required bet is more than the money you have left, the farmer must spend their **entire remaining balance** (going "all-in") for that round.
- Include your analysis within the notebook itself. You can use markdown cells for this.
- In addition to the analysis, include any observations you made about the results. You may also use plots to help with explanations.

Question 2 [30 Marks]

Consider a biased coin that comes up heads with probability p and tails with probability $1 - p$. Let q_n be the probability that after n independent tosses, there have been an odd number of heads.

Tasks:

1. Derive a recursion relating q_n to q_{n-1} .
2. Solve the recursion to obtain a closed-form expression for q_n .

Question 3 [30 Marks]

3.1

A hospital has two independent backup power generators. The hospital can continue operating as long as **at least one generator is working**. Each generator has a **3% chance of failing** during a 12-hour period.

Task:

What is the probability that the hospital will **lose power during the 12-hour period due to generator failure**?

3.2

In a roomful of 25 people, what is the probability that **at least two people have the same birthday**? Assume birthdays are **uniformly distributed** across 365 days and there is **no leap year complication**.

Task:

Compute the required probability.

Submission Format

- Create a folder named **RollNumber_A4-1** containing:
 - `q1.ipynb` (Code, Results and Analysis)
 - `q2_3.pdf`
- Zip the folder (Example: `2024101090_A4-A.zip`) for submission.

For Q1, ensure that you run the notebook before submitting so that all outputs are saved. Python file submissions are NOT allowed.

Only Pen-Paper handwritten submissions are allowed. Digitally handwritten submissions are NOT allowed.

Please start the assignment early to prevent clash with end sem preparation. Plagiarism is strictly prohibited and a straight 0 will be awarded in such cases.